# 21CSC529T Inferential Statistics

## Unit 1 : Introduction to Statistics

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# Syllabus :

Unit - 1: INTRODUCTION TO STATISTICS
Unit - 2: PROBABILITY
Unit - 3: ESTIMATION & HYPOTHESIS TESTING
Unit - 4: HYPOTHESIS TESTING -I
Unit - 5: HYPOTHESIS TESTING -II

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# Unit 1

- Role of statistics in Data science
- Different types of data. - Understanding Random variable, Numerical Variable and Categorical Variable
- Data Collection, Types and its formats
- Types of Sampling
- Descriptive Statistics - Measure of Central Tendency Mean, Median and Mode
- Measure of Dispersion Range, Quartiles, Standard Deviation, Variance
- Distribution of Data Skewness and Kurtosis
- Covariance and Correlation, Difference between covariance and correlation and its significance.

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# Agenda

- Introduction to Statistics
- Classification of Data
- Various Types of Sources
- Characteristics of Statistics
- Sampling and its types
- Descriptive Statistics
    - Measure of Central Tendency

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# ‘STATISTICS’

*“Statistics is not a body of substantive knowledge, but a body of methods obtaining knowledge”*

*-W.A.Wallis*

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# Statistics

- Thought to have been derived from the Latin word *'statisticum collegium'* meaning council of state
- The definition was limited to the collection of economic and demographic data
- In the 19th century the definition was broadened and included collection, summarization, and analysis of data
- <u>Definition:</u> *Statistics is the branch of mathematics that deals with the collection, analysis, interpretation, presentation, and organization of data. It provides tools and methods for making sense of numerical data, allowing researchers and analysts to draw conclusions, make predictions, and inform decisions.*

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# Data In - Sights

"Data" has become a buzzing theme in today's world.

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# What is Data

Data is units of information in structured or unstructured format

Examples:

- Collection of relevant tweets
- Records of yield in a farm over a period of time
- Records of stock price every minute
- Records of performance of a sports person

**“Collection of Facts”**

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# Kinds of Data?

- Sound
- Video
- Single character
- Number (integer or floating-point)
- Picture
- Boolean (true or false)
- Text (string)

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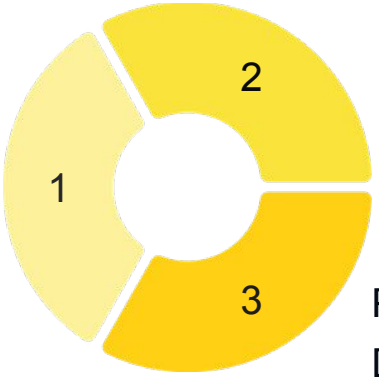
# Types of Data

## Quantitative Data

Numerical and measurable data such as height, weight, or temperature.




## Qualitative Data

Describes characteristics that cannot be measured numerically, like colors and labels.

## Primary vs. Secondary Data

Primary data is firsthand, while secondary data is collected from existing sources.

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# Sources of Data Collection

## Primary Sources

- Data collected by user
- Raw data gathered first-hand from the source
- Objective for data collection is pre-defined

## Secondary Sources

- Data is not collected by the user
- Data collected by someone else and is readily available
- Objective for the data collection could be different than the objective of the user

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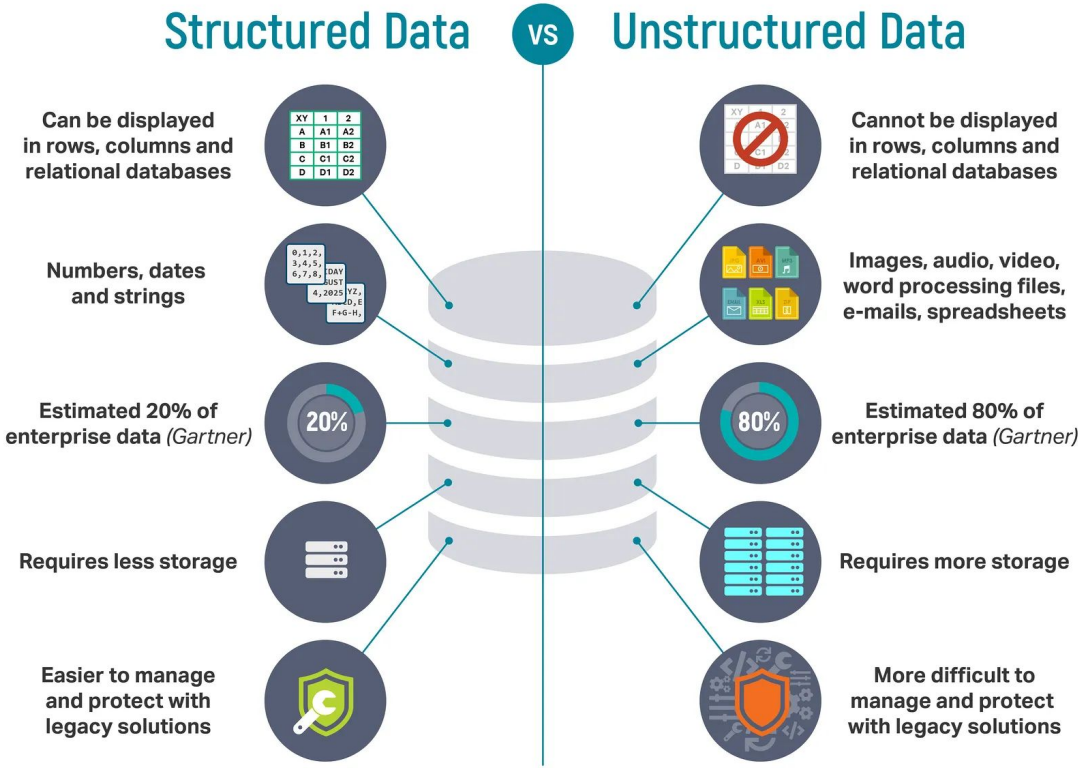
# Difference between structured Vs Unstructured Data

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# How to store Data?

- Data can be stored in temporary storage spaces called variables.
- A variable is a value that can change, depending on conditions or on information passed to the program.

## How to work with variables?

- Create Variables
- Storing the values
- Retrieving and using the stored value

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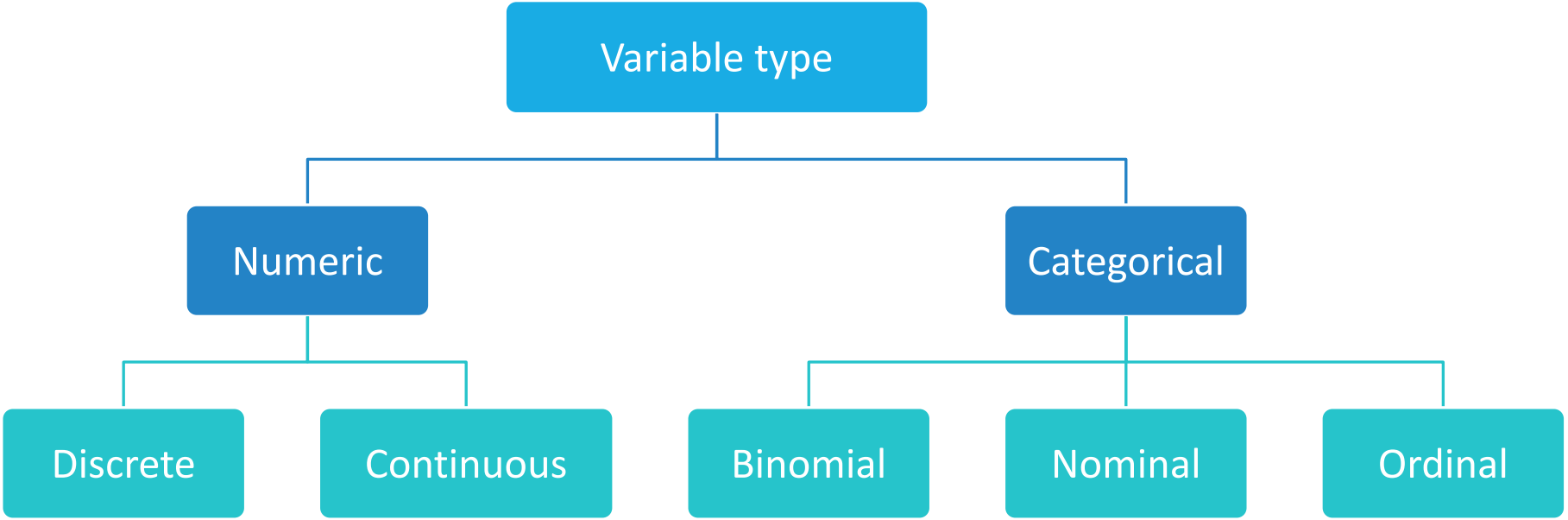
# Understand the variable

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# Discrete Numerical Variable

- A variable which takes a numeric value is called a numeric variable
- Also known as quantitative variable
- A discrete numeric variable is a random variable which takes discrete values, i.e. values from the set of whole numbers only. It can take countably finite values

**Examples:**

1. Number of cars passing by a toll-gate every minute
2. Number of defective items in a box

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# Continuous Numerical Variable

A continuous numeric variable is a variable which can have infinite number of values within a range

**Examples:**

1. The amount of rainfall in millimeters
2. Price of a stock

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# Categorical Variable

- Categorical variable has two or more levels
- Also known as qualitative variable

**Examples:**

1. Colors: red, orange, yellow
2. Gender: male, female
3. Economic status: low, medium, high

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# Categorical Variable

Nominal Data:

- Nominal data has no order and has two or more categories
- It represents discrete units and are used to label variables
- Example: Housing type - Apartment, Bungalow, Penthouse

Binary Data:

- Binary data has no order and has strictly two categories
- Also known as dichotomous variable
- Example: Presence or absence of a disease

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# Categorical Variable

Ordinal Data:

- Ordinal data is ordered nominal data
- It represents discrete and ordered units
- Example: Education - Primary, Secondary, High School, College and University which are
  ordered

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PLEASE!! NOTE

- At times data may appear to be numeric but is actually categorical

- Consider the adjoining table. The ‘Gender’ column has value 1  and 0

| Name | Gender | Age |
|---|---|---|
| Ria | 1 | 45 |
| Arya | 1 | 23 |
| Sam | 0 | 34 |
| Joe | 0 | 54 |
| Harry | 0 | 12 |
| Sarah | 1 | 43 |

- The numeric values used to represent a categorical variable cannot be used for numerical computation

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# Classification of Data

In general, there are four types of data. They are:

- Time Series Data
- Cross sectional data
- Pooled Data
- Panel Data

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# Time series data

- Time series data is the set of observations on a variable at different time points

- The data may be collected daily, weekly, monthly, or annually

| Date | Prev.Close | Open.Price | High.Price | Low.Price |
|---|---|---|---|---|
| 01-Jan-1996 | 408 | 407 | 407.9 | 405 |
| 02-Jan-1996 | 407.9 | 407 | 409 | 406.25 |
| 03-Jan-1996 | 406.25 | 409 | 409 | 409 |
| 04-Jan-1996 | 409 | 405 | 407 | 405 |
| 05-Jan-1996 | 406.3 | 401.5 | 401.5 | 401.5 |
| 08-Jan-1996 | 401.5 | 401.5 | 405 | 402 |
| 09-Jan-1996 | 404 | 404 | 400 | 395 |
| 10-Jan-1996 | 399.5 | 399.5 | 397 | 395 |
| 11-Jan-1996 | 396 | 399 | 405 | 396 |
| 12-Jan-1996 | 405 | 403 | 403 | 400 |

Sample data of daily stock price of a company

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# Cross-sectional data

Cross-sectional data are set of observations on two or more variables at the same time point

| Produce in the 2019 in (quintals) | | | |
|---|---|---|---|
| **Region** | **Rice** | **Wheat** | **Maize** |
| **Region 1** | 28.76 | 86.66 | 8.23 |
| **Region 2** | 84.03 | 65.45 | 41.94 |
| **Region 3** | 71.75 | 28.99 | 38.43 |
| **Region 4** | 14 | 68.87 | 7.04 |
| **Region 5** | 43.97 | 54.27 | 86.51 |

Sample data of a farm produce in five regions in the year 2019

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# Pooled data

Pooled data is combination of both time series data and cross sectional data.

| | Produce in the 2011 | | Produce in the 2012 | |
|---|---|---|---|---|
| Appliance | Production | Profit (in '000 Rs) | Production | Profit (in '000 Rs) |
| Radio | 280 | 13 | 121 | 8 |
| Television | 840 | 645 | 1456 | 5192 |
| Washing Machine | 775 | 2899 | 152 | 5706 |
| Computer | 876 | 6887 | 1005 | 6349 |
| Air Conditioner | 439 | 5427 | 328 | 8095 |

Sample data of the number of appliances produced in an industry for two years

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# Panel data

- Panel data is type of pooled data
- The same cross-sectional unit is surveyed over time
- Also known as longitudinal or micro-panel data

| Appliance | Year | Production | Profit (in '000 Rs) |
|---|---|---|---|
| Radio | 2010 | 280 | 13 |
| Television | 2010 | 840 | 645 |
| Washing Machine | 2010 | 775 | 2899 |
| Computer | 2010 | 876 | 6887 |
| Air Conditioner | 2010 | 439 | 5427 |
| Radio | 2011 | 234 | 123 |
| Television | 2011 | 835 | 645 |
| Washing Machine | 2011 | 564 | 2899 |
| Computer | 2011 | 874 | 6887 |
| Air Conditioner | 2011 | 435 | 5427 |

Sample data of the number of appliances produced and profits earned by a company over a period of two years

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# The real STATISTICS?

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# Which is real STATISTICS?

- Arjun secured 100/100 in Mathematics

- The average mark of students in the Mathematics course is 65

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# The reality?

- Statistics always involve numerical data, but not every piece of numerical data qualifies as statistics.
- For instance, the statement "Arjun secured 100/100 in Mathematics" is simply a numerical fact, not a statistical measure.
- Conversely, "The average mark of students in the Mathematics course is 65" represents a statistical idea because it summarizes data from a group.

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# Characteristics of Statistics

1. Aggregate of facts
2. Affected by several causes
3. Numerically expressed
4. Enumerated or estimated as accurately as possible
5. Collected in a systematic manner
6. Collected for a planned purpose
7. Comparability

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## 1. Aggregate of Facts

- "Aggregate" - Cannot be a single numerical value
- "Facts" - Only something that can be measured
- Examples
  - Height of a student is 5 feet 4 inches ✗
  - Heights of students of a certain class ✓

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## 2. Affected by Several Causes

- A statistic is affected by not a single but multiple causes
- This implies that we cannot study the effect of a particular factor in isolation
- Example
  - Sales of the latest model of an iPhone
  - Can be affected by the number of new features, price, supply, marketing, etc.

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## 3. Numerically Expressed

- A statistic should take a quantitative form and expressed as a numerical figure
- This implies that qualitative phenomenon cannot be statistics
- Example
  - Happiness of employees in an organization ✗
  - Number of employees in an organization who say they are happy ✓

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# 4. Enumerated or Estimated as Accurately as Possible

- Data collected by explicit counting- this will be exact
  - For example, the height of students in a class
  - Cannot leave out a single student
- Data collected by estimation when counting is not possible- accuracy varies
  - For example, average height of children aged 12-15 years in India
  - We cannot count every child in the age group thus we estimate using samples
  - Leaving out 10 children in the age group will not significantly affect the estimate

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# 5. Collected in a Systematic Manner

- Unmethodical and disorganized data collection is undesirable
- It hampers the accuracy of the measurement
- It may lead to us making wrong conclusions

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## 6. Collected for a Planned Purpose

- Before starting the process of data collection, the purpose should be clearly defined
- Resources and time will be wasted in collecting irrelevant data
- The relevant data may not be collected at all or its quality may be affected
- Example
  - We want to measure the number of Non-fiction books sold by a bookstore
  - Counting the number of fiction and comic books is irrelevant

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## 7. Comparability

- Being able to compare different statistics or over is what makes them useful

- Comparability is essential for making inferences, establishing or detecting changes in trends

- Example
  - We measure the heights of students from two different classes
  - We can then compare the average height of each of the two

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# Functions of Statistics

1. Concrete Factual Presentations
2. Reduce complexity of raw data
3. Make comparison and draw conclusions
4. Helps in forming Hypothesis
5. Making Predictions
6. Useful for Decision Making

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# Limitations of Statistics

1. Limited to quantitative study
2. Aggregate Measurements
3. Homogeneity of data
4. Results are true only on  Estimates and not accurate numbers
5. Leads to erroneous conclusions

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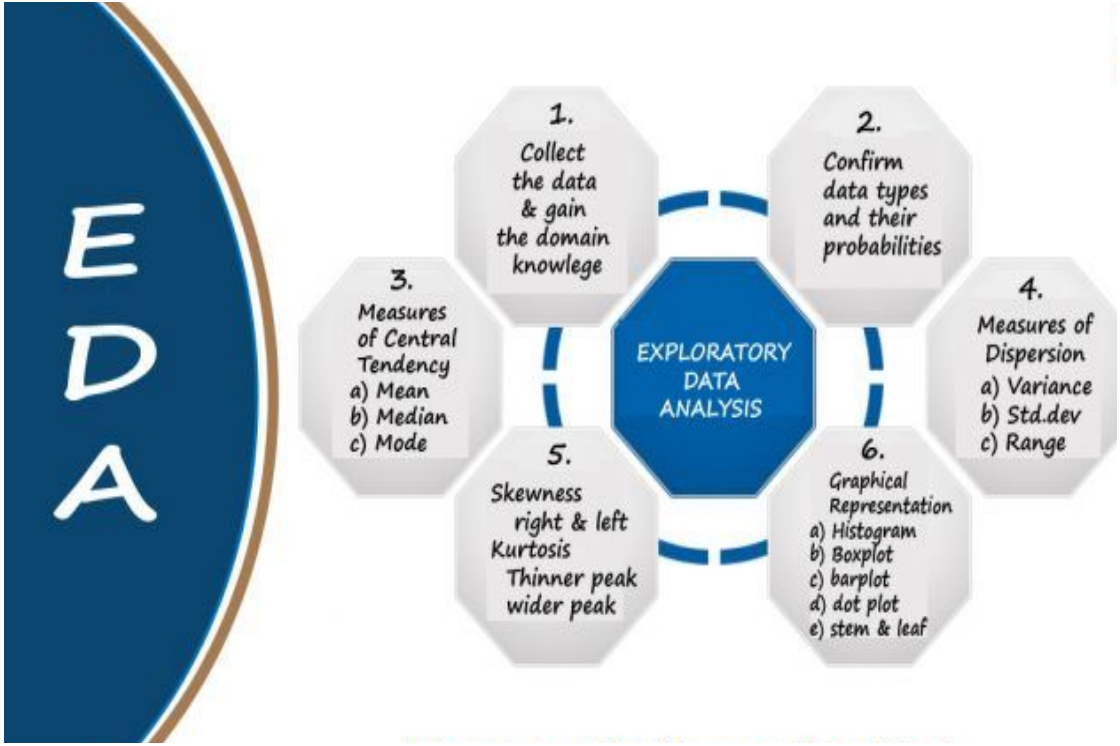
# Exploratory Data Analysis




- Understanding the data
- Identify Patterns and Relationships
- Generate Hypotheses
- Check Assumptions
- Prepare for Modeling

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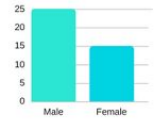
# Data Analysis

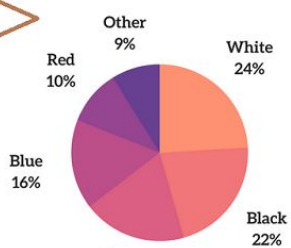
## Descriptive Analysis

- It is used to only describe the sample or summarize information about the sample
- Also known as Descriptive Statistics

| | A | B | C | D |
|---|---|---|---|---|
| 1 | Respondent Number | Age | Gender | Favorite Car Color |
| 2 | 1 | 22 | M | White |
| 3 | 2 | 37 | F | Silver |
| 4 | 3 | 45 | F | Black |
| 5 | 4 | 62 | F | Gray |
| 6 | 5 | 28 | M | Red |
| 7 | 6 | 45 | M | Green |
| 8 | 7 | 88 | F | Brown |
| 9 | 8 | 61 | M | White |
| 10 | 9 | 95 | M | Black |
| 11 | 10 | 27 | M | White |
| 12 | 11 | 39 | F | Green |
| 13 | 12 | 43 | M | Brown |
| 14 | 13 | 55 | F | Black |
| 15 | 14 | 59 | F | White |

**RAW DATA**

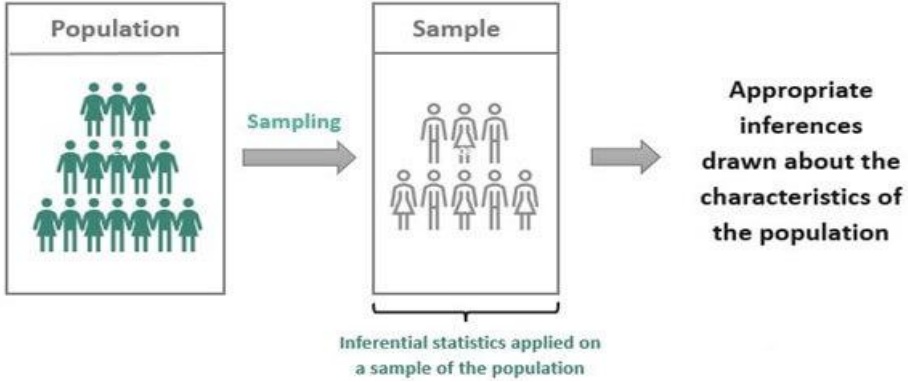



**Descriptive Statistics**

## Inferential Analysis

- It is used to make inferences and generalizations about the broader population.
- Also know an Inferential Statistics

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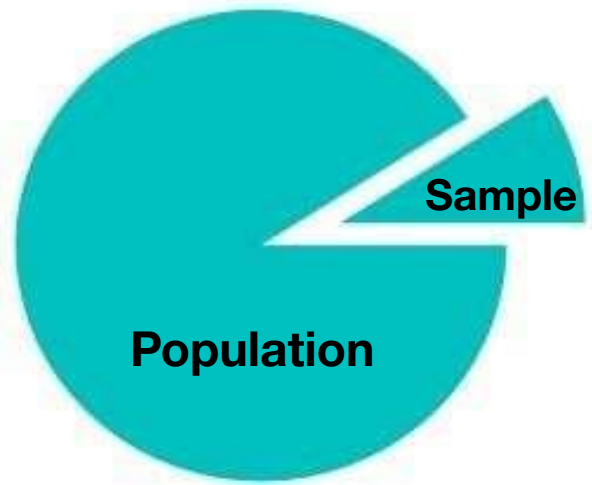
# Population & Sample

- **Population** is a collection of all the individuals or objects

- **Sample** is a subset of the population which is a representative of the population




## Types of Sampling

- Random Sampling
- Systematic Sampling
- Stratified Sampling
- Cluster Sampling
- Convenience Sampling

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# Sampling

- **Sampling** is the process of selecting a subset of individuals, items, or observations from a larger population to make inferences or predictions about the entire population.

- The goal of sampling is to gather data in a way that accurately reflects the population as a whole

- Sampling  save time, effort, and resources compared to studying the entire population.

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# Need to draw a sample

Suppose a company producing electric bulbs wants to know the average life of a bulb. If the details of all the bulbs are available, then this information is regarded as the population.

It would be challenging for the company to test each and every bulb produced. In such a scenario, the company would draw a sample from the produced bulbs to test.

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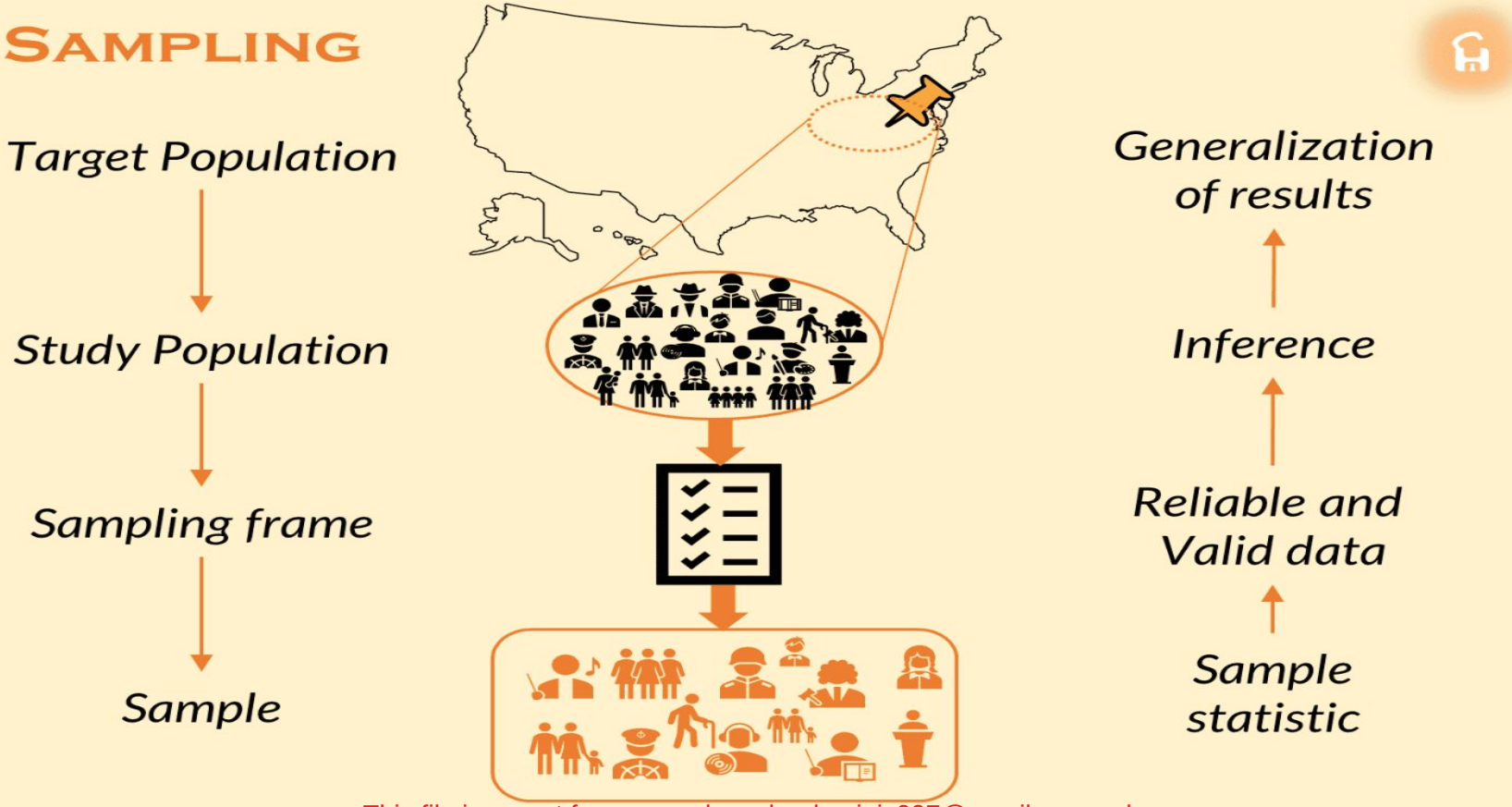
# SAMPLING

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Reference Link : https://conceptshacked.com/probability-sampling/

# Significance of Sampling

1. Resource Optimisation
2. Improved Data Accuracy
3. Facilitates Generalization
4. Ethical Considerations
5. Enhances Feasibility

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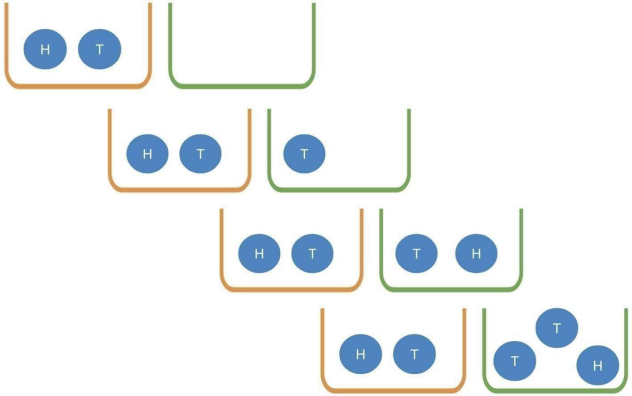
# Sample with replacement

The unit selected to be in the sample is returned to the population.

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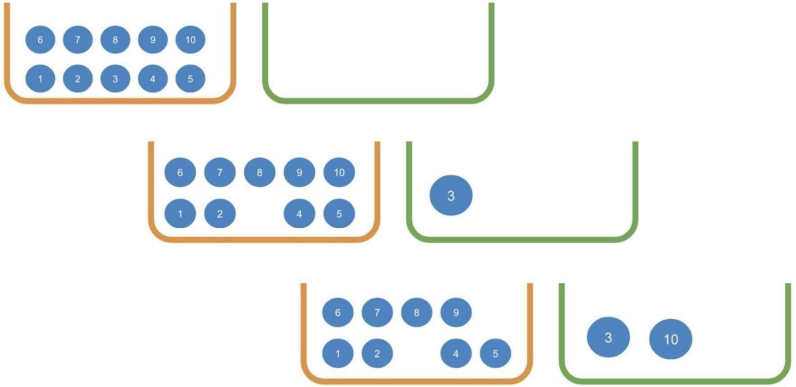
# Sample without replacement

The unit selected to be in the sample is not returned to the population.

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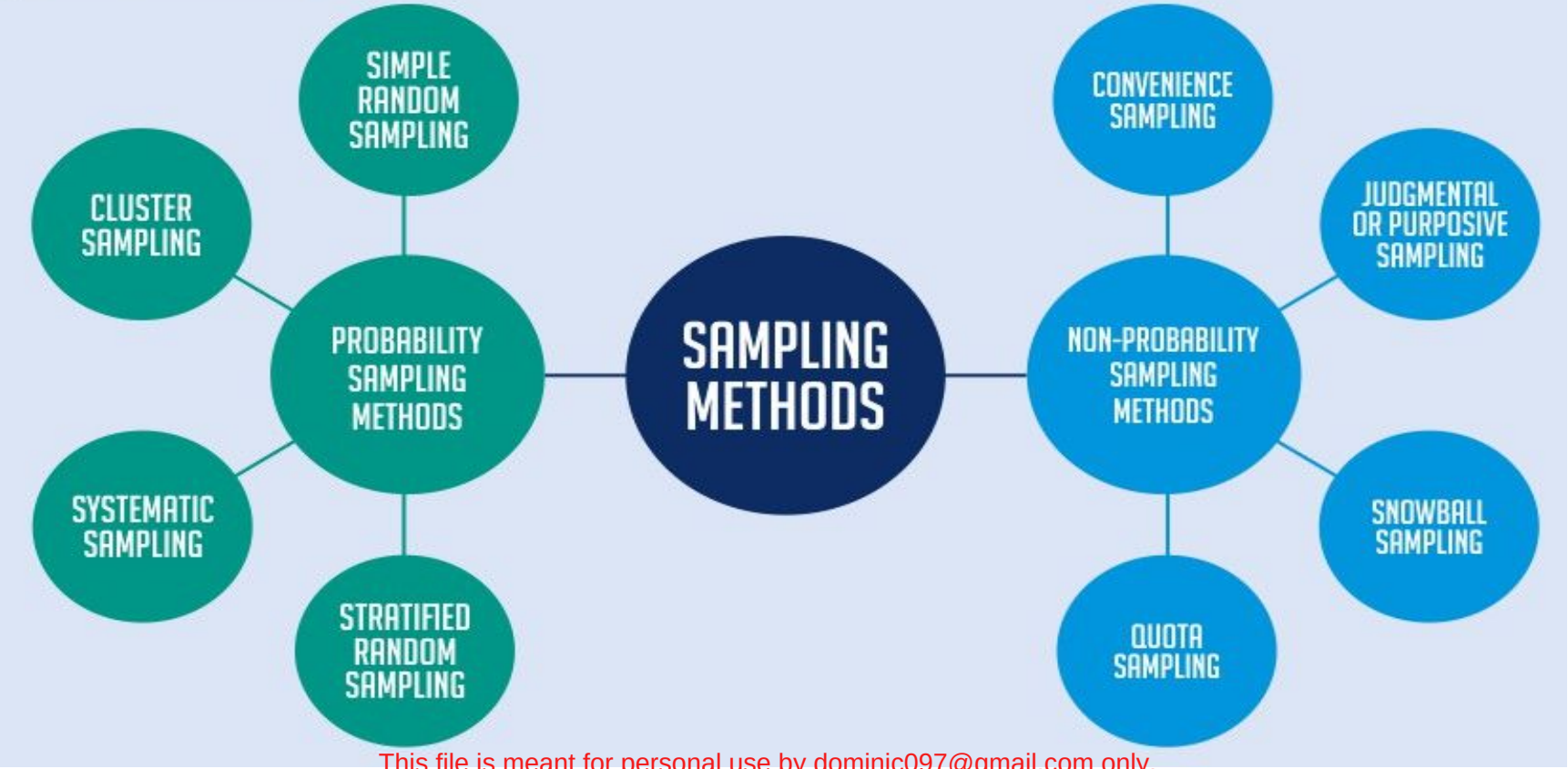
# Types of Sampling:

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Reference Link: https://www.questionpro.com/blog/types-of-sampling-for-social-research/

# Probability Sampling

- Probability sampling is a sampling technique where a researcher selects a few criteria and chooses members of a population randomly.

- All the members have an equal opportunity to participate in the sample with this selection parameter.

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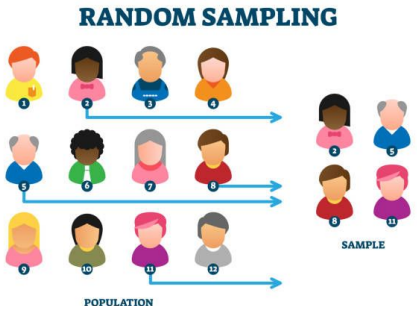
# Types of Probability Sampling

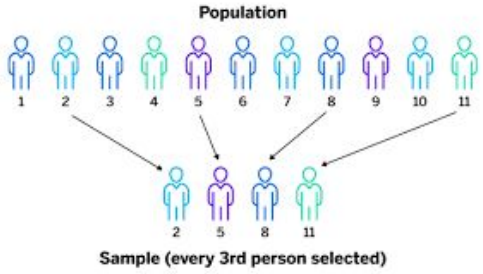
## Simple Random Sampling:

- Every member of the population has an equal chance of being selected.
- Respects the principle of independence
- Example: random sample of 5 students from a class of 50 students



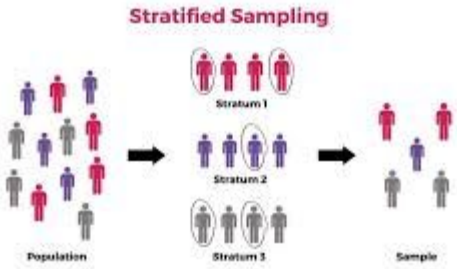

## Systematic Sampling:

- Selects every $k^{th}$ individual from a list or sequence.
- Example: Surveying every $3^{rd}$ person entering a store.




## Stratified Sampling:

- The population is divided into strata (groups) based on a characteristic (e.g., age, gender), and a random sample is taken from each stratum.
- In Stratified sampling we assume strata is homogenous.
- Example: Ensuring representation of different age groups in a survey.

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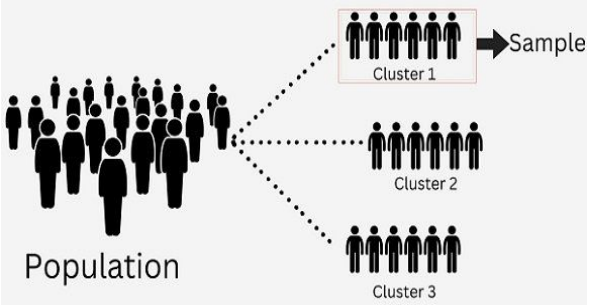
# Types of Probability Sampling

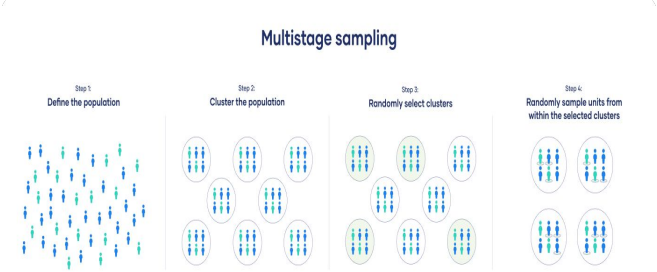
## Cluster Sampling:

- The population is divided into clusters (e.g., geographical areas), and a random sample of clusters is selected, with all individuals within those clusters surveyed.
- Naturally formed group with diverse samples with in group.
- Example: Selecting certain schools and surveying all students within those schools.




## Multistage Sampling:

- Combines several sampling methods, typically involving selecting samples in stages, e.g., selecting clusters first, then using simple random sampling within each cluster.
- Example: First selecting cities, then schools within those cities, and then students within those schools.

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# Non-Probability Sampling:

- In non-probability sampling, not every member of the population has a chance of being included, and the selection is often based on convenience or judgment rather than randomness.

- This method is often used when probability sampling is impractical.

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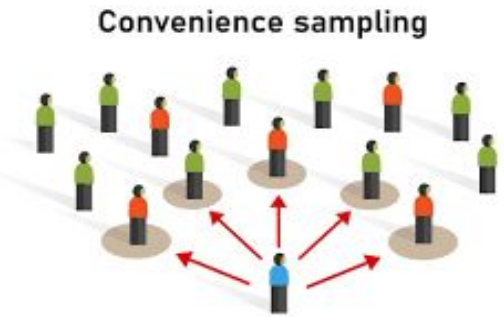
# Types of Non-Probability Sampling:

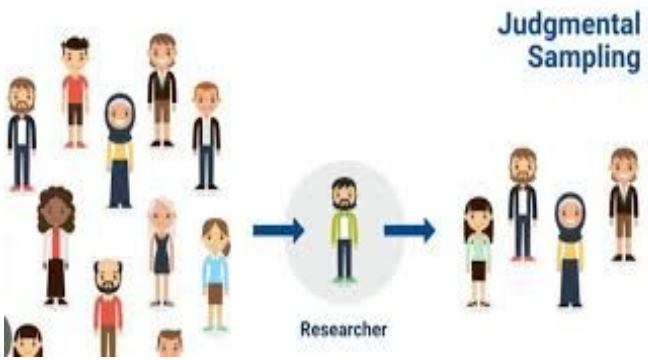
Convenience Sampling:

- The sample is taken from a group that is easy to access.
- Example: Surveying people at a mall because it's convenient.




Judgmental or Purposive Sampling:

- The researcher selects the sample based on their judgment of what is most representative or typical.
- Example: Choosing experts in a field for a study.

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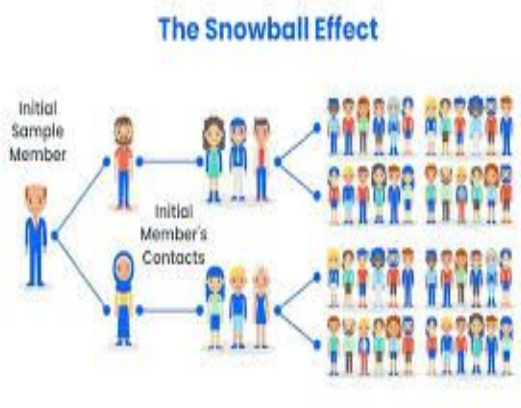
# Types of Non-Probability Sampling:

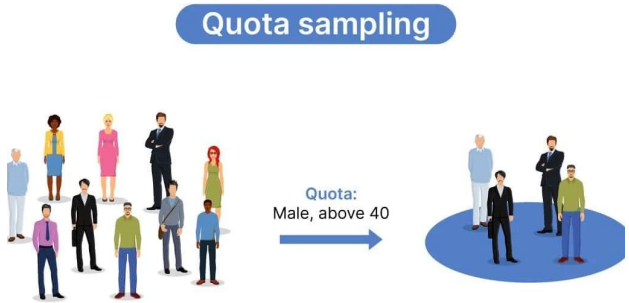
Snowball Sampling:

- Existing study subjects recruit future subjects from among their acquaintances.
- Example: A study on a rare disease where current participants refer others with the same condition.




Quota Sampling:

- The researcher ensures that the sample reflects certain characteristics of the population, but selection within those quotas is non-random.
- Example: Ensuring a sample includes a specific percentage of males and females, but selecting individuals conveniently.

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# Types of Sampling Error

- Random error

  Random errors are unsystematic, unpredictable, and just as likely to over-as under-estimate the “true” score.

  Difference between sample mean and true mean

  Can be reduced by selecting large sample size

- Non - Random error

  Non-random errors are systematic in over or under-estimating the true score

  It is error in sampling process while collecting data

  1. Sample bias
  2. Completion rate
  3. Interviewer effect

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